

Aayush KC

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Summary

Experienced Java Developer with 6+ years of expertise in building scalable, secure, and high-performance backend systems using Java, Spring Boot, and Microservices architecture. Proficient in RESTful API development, cloud deployment on AWS, and database management (SQL/NoSQL). Skilled in test automation (JUnit, Mockito, Cucumber), secure authentication (Spring Security, OAuth 2.0), and containerization with Docker and Kubernetes. Known for delivering reliable solutions in Agile environments and continuously adapting to emerging technologies like Java 17 and Kafka.

Skills

- Languages Java 8, 11, 17, J2EE, JavaScript, PL/SQL, SQL
 - Amazon Web Services
- Strong Interpersonal and Communication Skills
 - Linux Environments
 - Web Programming
- Frameworks Hibernate, Spring, Struts
- Web Technologies JSP, Servlets, JDBC, HTML5, DHTML, Custom Tags, JMS, JNDI, Java Mail, XML, CSS3, SOAP, WSDL, REST, AWS, SVN, GIT
 - App/Web Servers WebLogic, WebSphere, Tomcat
 - Tools Ant, Maven, Log4J, Eclipse, NetBeans, J-Unit
- Databases Oracle, MS SQL Server, MongoDB, Google Cloud Platform (GCP).
- Data Structures and Algorithms
- API Design and Development
- Design Patterns and Principles
- Software Components and Libraries
 - Agile/Scrum Methodology
- Test Driven Development (TDD).
 - Pair Programming
- Software Development Standards
 - Data Storage and Retrieval
 - Teamwork and Collaboration
- Multithreaded Software Applications
 - Agile, Jira

Experience

Capital One, Dallas, TX — Sr Software Developer

June 2024– Present

- Delivered high-performance backend systems using Java 17, adopting modern features like sealed classes, pattern matching, and records for robust and maintainable codebases.
- Led the migration of legacy systems from Java 8/11 to Java 17, utilizing enhanced switch expressions, sealed types, and API improvements to modernize applications.
- Applied Text Blocks in Java 17 to streamline the formatting and readability of complex JSON, SQL, and multi-line strings.
- Tuned low-latency applications using Z Garbage Collector (ZGC), improving system responsiveness and efficient memory usage.
- Enabled advanced application observability and debugging using Java Flight Recorder (JFR) and contextualized logging strategies.
- Created rich, modular frontends with Angular 17, building dynamic user interfaces with reusable components.
- Contributed to web app development using Python, gaining practical exposure in scripting, debugging, reporting, and API development.
- Built and containerized Spring Boot and Node.js services using Docker, and managed container orchestration through Kubernetes.
- Implemented Kafka producers and consumers on Kafka-Zookeeper clusters, handling message reprocessing by leveraging offset IDs for fault-tolerance.
- Designed microservices using JPA, Hibernate, and Spring Data, optimizing data interaction across enterprise systems.
- Developed scalable Angular modules using TypeScript, enhancing structure and maintainability across UI layers.
- Developed and maintained Angular 17 applications with TypeScript and NgRx state management, integrating with Java Spring Boot RESTful APIs.
- Implemented Redux pattern using NgRx for scalable and predictable state management in enterprise-level Angular applications consuming Java-based microservices.
- Created reusable Angular components, services, and modules following modern UI/UX patterns and corporate design systems.
- Architected and maintained Spring Batch pipelines to handle bulk data extraction from formats like XML, JSON, and CSV.
- Built reliable and scalable batch jobs with retry, skip logic, and custom listeners for logging, audit tracking, and failure notifications.
- Enabled restartable jobs using Spring Batch's job repository and execution context for resilience in case of partial failures.
- Administered MongoDB instances on AWS VPC, designing performant NoSQL access layers for modern applications.
- Deployed applications on AWS leveraging services such as EC2, RDS, S3, DynamoDB, SNS, SQS, and IAM, ensuring fault-tolerant, scalable architectures using CloudFormation templates.
- Developed and deployed Java microservices as part of Cloud App Factory initiatives focused on AWS cloud migration.
- Automated comprehensive testing using JUnit, Mockito, and Cucumber, significantly reducing regression and manual test overhead.
- Employed Terraform, Ansible, and Jenkins to automate infrastructure provisioning, configuration, and CI/CD workflows.
- Validated RESTful APIs using Postman, with collections automated using Newman CLI and integrated into CI pipelines for continuous assurance.
- Evaluated and validated API payloads, response codes, and SLA metrics in microservices environments.
- Optimized data models and queries in MySQL, PostgreSQL, and Oracle, enhancing performance and storage efficiency.
- Installed, upgraded, tuned, and monitored PostgreSQL instances for production-grade deployments.
- Integrated legacy systems into cloud-native ecosystems using MuleSoft Java connectors.

- Managed infrastructure-as-code deployments and Scala-based applications on AWS, configured using Ansible
- Ensured standardized API documentation and onboarding using Swagger and OpenAPI specifications.
- Investigated and resolved message-processing issues in Kafka using Kibana dashboards and Elasticsearch queries.
- Leveraged AWS and Kubernetes to build highly available, auto-scaling containerized applications for enterprise-grade environments.

Charles Schwab, Dallas, TX — Full Stack Java Developer

Jan 2022- June2024

- Designed and implemented microservices using Spring Boot and Spring Cloud, enhancing scalability, fault tolerance, and performance.
- Developed and maintained microservices with JPA, Hibernate, and Spring Data for efficient data handling.
- Applied pair programming to boost productivity, code quality, and knowledge sharing.
- Implemented service discovery using Netflix Eureka and Spring Cloud Consul for seamless microservices communication.
- Optimized application cost and performance on AWS, selecting appropriate services based on requirements.
- Administered databases on AWS, integrating a with other infrastructures and using services like EC2, S3, EBS, CloudFront, IAM, CloudFormation, and DynamoDB.
- Developed enterprise applications with Oracle, utilizing advanced SQL queries, PL/SQL, stored procedures, and triggers for efficient data manipulation.
- Created reusable components, custom modules, utility functions, services, and custom hooks in React and JavaScript.
- Built React services to interact with backend systems via REST API calls.
- Integrated React with Spring MVC, developed custom React hooks, and utilized Spring forms for seamless frontend-backend connectivity.
- Designed and developed a Spring Batch job to automate employee course deadline notifications, processing user enrollment data in scheduled batch executions.
- Implemented custom components to load enrollment data, evaluate upcoming deadlines and dispatch email alerts.
- Scheduled Spring Batch jobs using Spring Scheduler and Quartz, ensuring timely and repeatable execution for deadline-based notifications.
- Optimized job execution performance using partitioning and multi-threaded step execution in Spring Batch.
- Built chunk-oriented batch processing to efficiently handle large datasets, enabling parallel processing for high throughput.
- Configured fault-tolerant batch steps with retry and skip logic, preventing failure due to transient email service or data issues.
- Validated forms using React and regular expressions, ensuring both client-side and server-side validation.
- Designed and developed applications using React, Node.js, MySQL, TypeScript, HTML5, CSS3, and Bootstrap to create modern, responsive user interfaces.
- Implemented Snowflake's unique features like virtual warehouses and automatic scaling to improve query performance, resulting in faster data analytics for the client.
- Worked on Converse application that managed the background process of call transcriptions and storage. It included lambdas and snowflakes.
- Integrated TDD into the development process and CI/CD pipelines using JUnit and Mockito for Java applications.
- Optimized database queries and schema design with Hibernate, improving system efficiency.
- Implemented secure REST APIs with Spring Security and OAuth 2.0, enabling seamless frontend-backend integration.
- Worked with Kafka to set up clusters and build real-time data streaming pipelines, providing real-time analytics.
- Designed a production-ready Kubernetes AWS infrastructure with microservice orchestration, ensuring high availability and scalability.
- Experience in setting up Confluent Schema Registry as a centralized service to manage Protobuf schemas.
- Used Kafka clients with the Schema Registry to serialize and deserialize data.
- Executed seamless application migrations to GitHub Actions, streamlining pre-CI tasks and enabling integration with Spinnaker pipelines for continuous delivery.
- Used GitHub as Version control for code repository and Jenkins for continuous integration.
- Utilized Swagger and Postman for API design, testing, and documentation, improving team collaboration.
- Managed AWS infrastructure deployment, monitoring, and configuration, migrating applications to shared pipelines for better production releases.
- Ensured application design allowed for flexible updates to Java classes and database access points without system disruption.
- Continuously updated skills in emerging technologies like Docker and Kubernetes to keep systems competitive.
- Extensively used JUnit and Mockito to test critical components such as database queries, REST APIs, and message queues.

Elevance Healthcare, Indianapolis, IN(Remote) — Java Developer

Nov 2020- Dec 2021

- Designed and developed user-friendly, responsive web interfaces using HTML5, CSS3, JavaScript, jQuery, and Angular, ensuring optimal performance across various devices.
- Implemented form validation using Angular and HTML5, enhancing data integrity and user experience.
- Developed RESTful microservices using Spring Boot and Spring MVC, ensuring modular and scalable architecture.
- Utilized Spring IoC for dependency injection, improving maintainability, and implemented Spring AOP for cross-cutting concerns such as logging, exception handling, and transaction management.
- Integrated Hibernate for ORM and optimized database interactions with Spring Data JPA, improving data retrieval performance.
- Developed data models and entity relationships, implementing Hibernate mapping and transaction management for efficient persistence.

- Designed and implemented REST APIs for seamless communication between backend services, utilizing JSON for data transformation and ensuring integration with external systems.
- Developed and deployed containerized applications using Docker, improving portability and scalability.
- Configured and deployed applications on GCP ensuring high availability and automated scaling.
- Integrated Spring DAO and Hibernate to manage database transactions efficiently for scalable data storage.
- Implemented message-driven architecture using JMS and ActiveMQ, ensuring reliable and asynchronous communication between services.
- Used Kafka to stream data and used consumers to consume the data from Kafka.
- Reprocessed failed Kafka messages using offset IDs and implemented Kafka producers and consumers on a Kafka-Zookeeper cluster.
- Developed and managed CI/CD pipelines using Jenkins, automating the build, test, and deployment process.
- Monitored and optimized application performance using Log Analytics, and Application Insights, proactively identifying and resolving issues.
- Implemented API Gateway and security best practices using OAuth 2.0 and JWT authentication, securing microservices.
- Created and deployed microservices on the Google Cloud Platform (GCP) utilizing Cloud Run and Cloud Functions.
- Managed Kubernetes clusters on GCP, optimizing autoscaling and fault tolerance, reducing operational downtime.
- Contributed to open-source Python projects on GitHub, focusing on data processing and GCP workflow integrations.
- Expertise of using Cloud SQL and Cloud Spanner to manage databases on the Google Cloud Platform (GCP).
- Performed integration, regression, and unit testing using JUnit, Postman, and Mockito, ensuring robust and reliable application releases.
- Managed source control and versioning using GIT, ensuring collaborative and efficient development.
- Worked in an Agile/Scrum environment, actively participating in sprint planning, daily stand-ups, and retrospective meetings to align development efforts with business goals.

Education.

Bachelor's in Computer science, Youngstown State University, OH